MIL-C-17157G
4 June 1985

SUPERSEDING
MIL-C-17157F
29 May 1979

MILITARY SPECIFICATION

CLOTH, KNITTED, COTTON (WAFFLE TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of waffle knitted cotton cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1491 - Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 6.2 and 4.3).

3.3 Material. (see 6.5).

3.3.1 Cotton. The cotton shall be of suitable staple length and grade to conform to this specification.

*3.3.2 Yarn. The yarn shall be made from cotton which has been carded, combed and spun into nominal 10-14's count singles for circular knit machine or nominal 18-22's count singles for the flat bed machine when tested as specified in 4.6.

*3.3.3 Knitting machines. The knitting machine may be a two bar (needle and guide) raschel with 12 to 13 needles per inch (4.7 to 5.1 per cm) on each needle bar (24 gauge). Guide bars to be threaded 1 in and 1 out on both bars.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

Example:

Chain Setting for Links

	Front Guide F Bar	Back Guide B Bar
4 or 5 repeats	(4	(10
	(0	(6
	(2	(4
	(6	(8
	(10	(4
	(6	(0
	(4	(2
	(8	(6

NOTE: The cloth may be knitted on machines of different gauges or types providing the resulting fabric is of the desired characteristics and meets the physical requirements specified in this specification.

3.3.4 Knit. The waffle effect is to be produced by a series of broken ridges or depressions, in both the length and width direction, on both sides of the fabric in order to produce a dead air pocket construction. The number of repeats of the knitting pattern shall be a minimum of 18 and a maximum of 22 per 10 inches (25.cm) in the wale direction and a minimum of 65 and a maximum of 85 per 10 inches (25.4cm) in the course direction when tested as specified in 4.6.

3.3.5 Physical requirements. The physical requirements of the finished cloth shall be as shown in Table I when tested as specified in 4.6.

3.3.5.1 Dimensional stability. The finished cloth shall not shrink nor elongate more than 12.0 percent in either the wale or course direction after laundering when tested as specified in 4.6.

Table I - Physical requirements

Weight oz/sq yd (g/m^2)	Ball Bursting Strength lbs (min)(newton)	Thickness inch (min)(cm)	
		at 0.1 psi (689 Pa)	at 1.1 psi (7583 Pa)
11.1 ± 1.0 (376 ± 34)	100 (445)	0.08 (0.20)	0.06 (0.15)

3.3.6 Width. The overall minimum width of the finished fabric shall be as specified (see 6.2).

3.3.7 Color. The cloth shall be of a natural color (undyed and unbleached).

3.3.8 Finish. The cloth shall be scoured (not bleached) and thoroughly rinsed. The finished cloth shall be clean and have a soft, lofty hand.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

3.3.8.1 Non-fibrous materials. The finished cloth shall not exceed 3.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.6.

3.3.8.2 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.6.

3.4 Weight of individual pieces. Unless otherwise specified (see 6.2), each piece shall weigh not less than 25 pounds (11.4kg) nor more than 35 pounds (15.9kg). Each length shall be put-up in folded pieces as specified in PPP-P-1134.

3.5 Fiber identification. Each piece shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.6 Workmanship. The finished cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination. The required yardage of each piece shall be examined and visual defects as defined in MIL-STD-1491 shall be classified as listed. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each yard (.91m) length-wise or fraction thereof in which it occurs. The sample unit shall be one linear yard (.91m). The lot size shall be expressed in units of one (1) pound (.45kg) each. The acceptable quality levels shall be 4.0 defects per 100 units (yards). The sample size shall be in accordance with inspection level II of MIL-STD-105. The number of pieces from which the sample is to be selected shall be in accordance with Table II. If this number is inadequate to provide the yardage required for examination, sufficient additional pieces shall be selected for yard-by-yard examination. An approximate equal number of yards shall be examined in each piece sampled.

Defects
Hole, cut, tear, run or dropped stitch
Abrasion mark resulting in weak area
Tuck stitch, pulled stitch or needle streak
Mend (any) 1/
Uneven, ridgy, or wrinkled or puckered cloth
Broken or missing yarn, missing for more than 1/2 inch (1.3cm)
Coarse or fine yarn 1/
Misknit - disruption in normal knitting pattern
Number of repeats per inch (cm) vary from normal adjacent fabric
Foreign matter 1/
Permanent crease in cloth
Any knot extending above surface of cloth
Stretch (stitch tension) knitting tension too tight or too loose
Shade bar 1/
Any spot or stain 1/

1/ Clearly visible at normal inspection distance (approximately three feet (.91m).

4.4.2.2 Overall examination. Each defect listed below shall be counted not more than once in each piece examined. The unit of production for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the number of defects acceptable shall be as shown in Table II. The pieces for this examination shall be those selected for yard by yard examination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

Defects

Objectional odor
Overall uncleanness
Uneven knitting throughout the piece
Edges frayed, ravelled, curled, rolled, folded, scalloped or uneven continuously or intermittently throughout the piece
Color not natural, undyed and unbleached

Table II

Lot size (lbs) (weight)	Sample size (pieces)	Maximum number of defects acceptable in sample
1200 or less 1/	3	0
1201 up to and including 3,200	5	0
3,201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If lot contains less than three pieces, each piece in the lot shall be examined.

4.4.3 Examination for poundage (weight).

4.4.3.1 Individual pieces. The piece shall be examined for gross weight. Any gross weight found to be less than the specified minimum weight or more than the specified maximum weight, or any gross weight found to be more than 1 pound (.45kg) below the gross weight on the piece ticket shall be considered as a defect with respect to weight. The unit of product for the examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the number of defects shall be in accordance with Table II.

4.4.3.2 Total poundage (weight) in sample. The lot shall be unacceptable if the total of the actual gross poundage of pieces in the sample is less than the total gross poundage marked on the piece tickets. The pieces examined shall be those seleced for examination of individual pieces.

4.4.4 Examination for compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each piece not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.5 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134, to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

4.6 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table III shall be followed. All tests shall be conducted on specimens which have been conditioned in a relaxed, supported state prior to testing. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3 continuous yards, (2.7m) full width of the finished cloth. The lot size shall be expressed in units of one pound (.45kg) each. The lot shall be unacceptable if one or more units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with the following:

Lot size (pounds)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table III - Test Methods

Characteristic	Requirement paragraph	Test Method
Identification of cotton	3.3.2	1200 1/
Combed cotton yard	3.3.2	1/
Yarn size	3.3.2	4020 2/
Type of knit	3.3.4	Visual
Pattern repeats per 10 inches (25.4cm) (wales and courses)	3.3.4	Visual
Weight	Table I	5041
Bursting Strength	Table I	5120
Thickness	Table I	5030
Shrinkage or elongation in laundering	3.3.5.1	5556 3/
Non-fibrous materials	3.3.8.1	2611
pH	3.3.8.2	2811

1/ A certificate of compliance shall be submitted for this requirement.
2/ A failure with respect to yarn count shall exist when a unit is coarser or finer than stated in 3.3.2. One cone or spool of whatever type cylinder shall be submitted for testing of the yarn.
3/ The cotton laundering procedure of Test Method 5556 shall be followed. The sample shall be conditioned in a relaxed state for 4 hours before the shrinkage measurements are taken.

5. PACKAGING

5.1 Preservation-Packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The knitted fabric covered by this specification is intended for use in the manufacture of cold weather underwear (drawers and undershirts) worn by military personnel.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.
c. Width required (see 3.3.6).
d. Weight of individual pieces (see 3.4) if other than specified.
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 Recycled material. It is encouraged that recylcled material be used when practical as long as it meets the requirements of the specification.

Custodian:
Navy - NU
Air Force - 99

Review activity:
Air Force - 82
DLA-CT

Preparing activity:
Navy - NU

Project No. 8305-0926

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-17157G	CLOTH, KNITTED, COTTON (WAFFLE TYPE)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2026-10-18T09:48Z
Check the source to verify that this is the current version before use.